MIL-C-47246(MI)
26 July 1974
SUPERSEDING
MIS 16285A
22 February 1971

MILITARY SPECIFICATION

COATINGS, ABLATIVE, APPLICATION OF

This specification is approved for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements for the application of ablative coating primer and ablative coating.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Federal

O-T-620	Trichloroethane-1,1,1, Technical, Inhibited (Methyl Chloroform)
TT-M-261	Methyl-Ethyl-Ketone, Technical

Military

MIL-P-47215	Primer, Ablative Resistant Coating
MIL-C-47244	Coating, Ablative Resistant

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STANDARDS

Federal

FED-STD-406 Plastics, Methods of Testing

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Equipment.

3.1.1 Fabricating facility. The fabricating facility for application of the coating shall provide the following:

a. Protection of the cleaned surfaces and the coating materials from dust, dirt, and water.

b. Normal ambient conditions consisting of an atmospheric pressure between 28 and 32 inches of mercury, a temperature between plus 15 and plus 35 degrees Celsius (C) (plus 59 and plus 95 degrees Fahrenheit (F)) and a relative humidity not greater than 95 percent.

3.1.2 Balance. The balance shall be capable of maintaining accuracy within 1 percent over its entire range. The balance shall be calibrated and certified to the prescribed accuracy not less than once every 90 days.

3.1.3 Heat source. Any suitable heat source such as a heat gun, oven, hot plate, or application tooling containing heating elements may be used to heat the material.

3.1.4 Miscellaneous equipment. Miscellaneous equipment used in the mixing, stirring, application, or other handling of the coating materials shall be made from materials having no detrimental effect on the coating and visibly free of dust or other contamination. Use of metal spatulas, containers, and mixing equipment shall be suitable, but use of porous and absorbent materials such as paper and wood shall be avoided.

3.1.5 Tooling. The tooling, such as edge and grid strips, metal masks, and form templates, shall be capable of producing finished parts conforming to the requirements of this specification and related engin-

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eering drawings. The form templates shall be inspected for correct dimensions not less than once every 90 days.

3.2 Materials.

3.2.1 Direct.

3.2.1.1 Ablative coating. Ablative coating shall be as specified on the engineering drawing and shall conform to MIL-C-47244.

3.2.1.2 Ablative coating primer. Primer used with the ablative coating shall be as specified on the engineering drawing and shall conform to MIL-P-47215.

3.2.2 Indirect.

3.2.2.1 Abrasive paper. Abrasive paper shall be silicon carbide, garnet, or aluminum oxide, 80 to 180 grit.

3.2.2.2 Wire brush. Wire brush shall have bristles made from corrosion-resistant steel.

3.2.2.3 Cheesecloth. Cheesecloth shall be clean, white, and lint-free.

3.2.2.4 Masking tape. Masking tape shall be pressure-sensitive.

3.2.2.5 Mold release. The mode release shall be a nonblocking material for free release of fixtures. Silicone or other transferable materials shall not be used.

3.2.2.6 Cleaning solvent. Cleaning solvent shall be methyl-ethyl-ketone conforming to TT-M-261; trichloroethane-1,1,1 conforming to O-T-620; or equal.

3.3 Procedure.

3.3.1 Solvent-cleaning. The area to be coated (as specified on the engineering drawing) shall be solvent-cleaned as follows:

a. Remove grease, oil, or other contamination by wiping with cheesecloth (see 3.2.2.3) saturated with cleaning solvent (see 3.2.2.6).

b. Wipe dry with clean, dry cheesecloth.

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c. Allow to air-dry for not less than 15 minutes. Repeat 3.3.1a and b until all visible evidence of grease, oil, or other contamination is removed.

Warning: All safety regulations regarding the handling of flammable and toxic materials shall be observed.

3.3.2 Application of mold release. A thin, uniform coat of mold release (see 3.2.2.5) shall be applied over those areas of the tooling fixtures which will be in contact with the ablative coating.

3.3.3 Masking. Areas to remain uncoated shall be masked with metal or plastic masks or tape (see 3.2.2.4) as required.

3.3.4 Abrasion. All surfaces to be coated shall be abraded with abrasive paper (see 3.2.2.1), wire brush (see 3.2.2.2), or non-ferrous abrasive blasting until clean, bare metal is exposed and recleaned in accordance with 3.3.1.

3.3.5 Application of coating primer. Coating primer (see 3.2.1.2) shall be applied in a single coat by brushing, spraying, or wiping. The coating primer shall be allowed to dry for not less than 30 minutes prior to application of the ablative coating.

3.3.6 Application of ablative coating. The ablative coating (see 3.2.1.1) shall be applied as follows:

a. Install edge and grid strips as necessary to control the coating depth, using double-backed adhesive tape or clamps.

b. Thoroughly mix the two-component coating material (see MIL-C-47244) in the ratio of 515 parts by weight of base resin to 1000 parts by weight of curing agent.

Note: The application life of 15,000 grams of mixed coating is 2 hours at plus 24 degrees C (plus 75 degrees F). If the temperature is lowered, the application life will be lengthened.

c. Apply the coating over the printed areas, as specified on the applicable engineering drawing, by troweling. Work the coating into small areas, exercising care to prevent voids and air pockets and to maintain the coating thickness compatible with the depth control strips installed and the form

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templates used. Heat may be applied to the coating material of tooling (see 3.1.3) to facilitate material flow and improve finish smoothness but care shall be taken to avoid excessive reduction of material application life (see 6.1). Open any existing air pockets and fill voids with additional coating material.

d. At temperatures ranging from plus 16 to plus 38 degrees C (plus 60 to plus 100 degrees F), expose the coating for not less than 8 hours of curing.

e. Upon completion of the curing cycle, remove all edge and grid strips and miscellaneous tooling. The uncoated areas exposed by the removal of the grid strips shall then be processed in accordance with 3.3.1 through 3.3.5, 3.3.6c, and 3.3.6d.

3.3.7 Rework. Defective areas may be reworked by cleaning the area and applying additional primer and coating as required. Eroded areas may be reworked as follows:

a. Remove any charred ablative coating with abrasive paper or wire brush and solvent-clean as specified in 3.3.1.

b. Apply additional coating (see 3.2.1.1) as required.

c. Cure as specified in 3.3.6d.

Note: Use coating primer (see 3.2.1.2) whenever the erosion extends to the metal structure.

3.3.8 Coating hardness. The hardness of the coating, after curing as specified in 3.3.6d, shall be not less than 45 Shore D.

3.4 Workmanship. The coating shall conform to the requirements of the engineering drawing. The coating shall be continuous, uniform, free from foreign material, and free of excessive pits or blisters. Due to coating material viscosity, the entrapment of small air bubbles during application cannot be avoided; but pits or bubbles with a dimension in excess of 3/16-inch shall be reworked as specified in 3.3.8.

4. QUALITY ASSURANCE PROVISIONS

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4.1 Responsibilty for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Inspection records. Suitable inspection records substantiating that requirements of 3.1, 3.2, 3.3, and 3.4 have been complied with shall be maintained and shall be available to the procuring activity.

4.3 Acceptance testing. Each coated part shall be subjected to the visual inspection and coating hardness test specified herein. Failure to meet the test requirements shall be cause for part rejection.

4.3.1 Visual inspection. Unless otherwise specified by the procuring activity, each coated part shall be visually inspected 100 percent and shall conform to the requirements of 3.4. The thickness of coating may be verified by demonstrating compatibility with the depth control established by grid and edge strips and form templates used in the application techniques of 3.3.6.

4.3.2 Coating hardness test. The coating hardness shall be determined in accordance with FED-STD-406, Method 1083, and shall be as specified in 3.3.8.

5. PREPARATION FOR DELIVERY

This section is not applicable to this specification.

6. NOTES

6.1 Intended use. This specification is intended to be used to establish the requirements for the application of ablative coating primer and ablative coating.

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6.2 Ordering data.

a. Title, number, and date of this specification.

6.3 Application life reduction. Curing of the ablative coating material is rapidly accelerated by heat, with full cure occuring in approximately 30 minutes at plus 66 degrees C (plus 150 degrees F). Therefore, the use of heat in processing should be conditioned by the detrimental effects of reduced application life of the heated material.

6.4 Supersession data. This specification includes the requirements of Missile Interim Specification MIS-16285A, dated 22 February 1971.

Custodian:
ARMY-MI

Preparing Activity:
ARMY - MI
Project No. 8030-A058

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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